CAS SciFinder

Task History

Initiating Search

November 10, 2025, 4:20 PM

Search:

CAS Lexicon Search:

Transhalogenation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (579)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (282) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

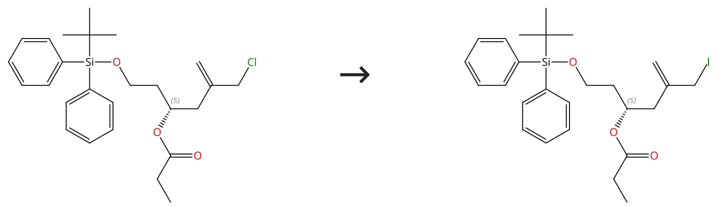
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-5796396

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

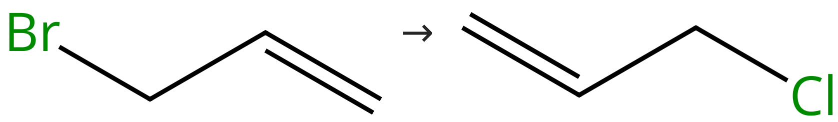
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (57)

Suppliers (55)

31-046-CAS-1427565

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

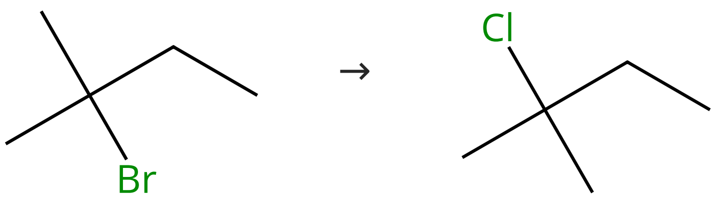
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (53)

Suppliers (43)

31-046-CAS-12696341

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

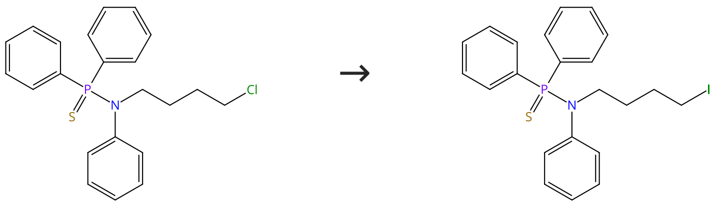
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 4 (1 Reaction)

Steps: **1** Yield: **100%**



31-046-CAS-14361396

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; reflux

Experimental Protocols

N-alkylation of amidothiophosphoryl compounds under phase-transfer catalysis conditions. Synthesis and properties of 1,3,2-thiazaphosphacyclanes

By: Artyushin, O. I.; et al

Russian Chemical Bulletin (2009), 58(1), 216-222.

Scheme 5 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (10)

31-046-CAS-84185

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bromide ($^{76}Br^{1-}$)
Solvents: Acetonitrile, Water; 100 °C

1.2 **Catalysts:** *trans*-Di(μ-acetato)bis[*o*-(di-*o*-tolylphosphino)benzyl]dipalladium
Solvents: Toluene; 40 min, 110 °C

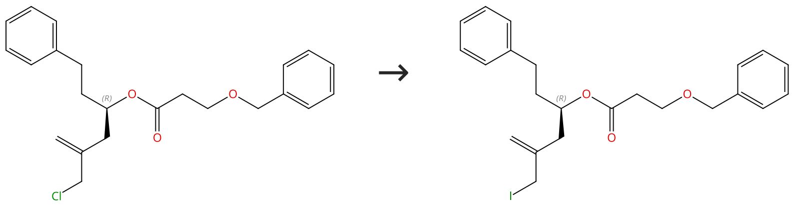
Radiobromination of closo-carboranes using palladium-catalyzed halogen exchange

By: Winberg, Karl Johan; et al

Journal of Labelled Compounds & Radiopharmaceuticals (2005), 48(3), 195-202.

Scheme 6 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-13126529

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

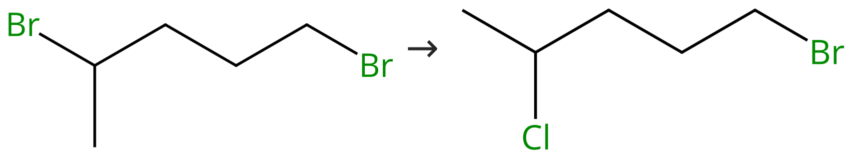
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 7 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (59)

Suppliers (7)

31-046-CAS-7824937	Steps: **1** Yield: **100%**
1.1	**Reagents:** Bismuth trichloride **Solvents:** 1,2-Dichloroethane

BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 8 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (5)

31-046-CAS-21792100	Steps: **1** Yield: **100%**
1.1	**Reagents:** Boron trichloride **Solvents:** Dichloromethane, Water; 20 min, rt

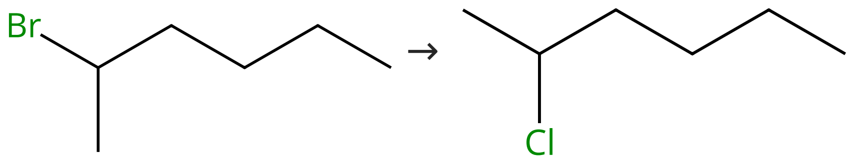
Synthesis, properties and reactivity of BCl_2 aza-BODIPY complexes and salts of the aza-dipyrrinato scaffold

By: Diaz-Rodriguez, Roberto M.; et al

Organic & Biomolecular Chemistry (2020), 18(11), 2139-2147.

Scheme 9 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (52)

Suppliers (50)

31-046-CAS-12411839	Steps: **1** Yield: **100%**
1.1	**Reagents:** Bismuth trichloride **Solvents:** 1,2-Dichloroethane

BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 10 (1 Reaction)

Steps: **1** Yield: **100%**

Absolute stereochemistry shown, Rotation (+) → Absolute stereochemistry shown, Rotation (+)

31-046-CAS-10378693 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

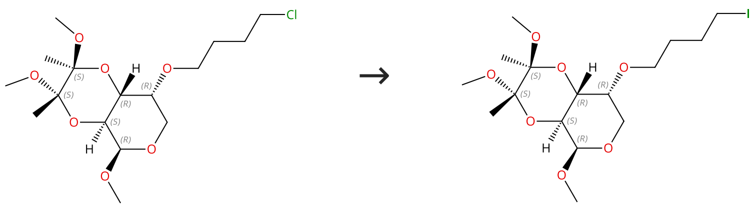
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 11 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (+) → Absolute stereochemistry shown, Rotation (+)

31-046-CAS-10853194 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, reflux

Experimental Protocols

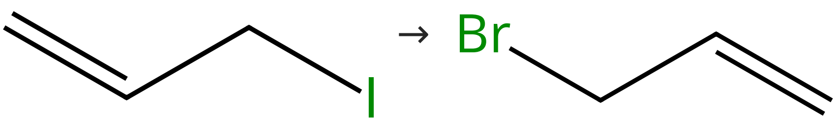
Synthesis and hydrolysis studies of novel glyco- functionalized platinum complexes

By: Moeker, Janina; et al

Carbohydrate Research (2012), 348, 14-26.

Scheme 12 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (47) Suppliers (57)

31-046-CAS-14830769 Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

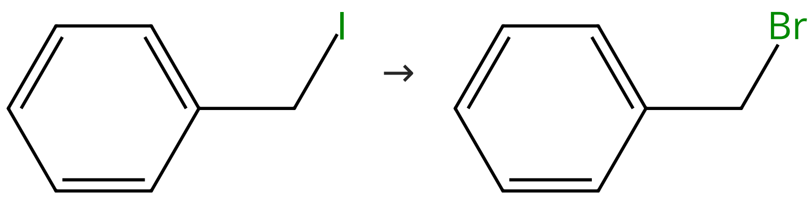
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (41)

Suppliers (67)

31-046-CAS-3572955

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

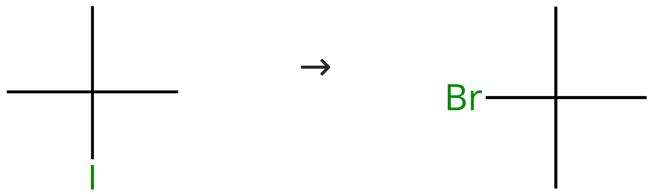
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (37)

Suppliers (60)

31-046-CAS-14553700

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

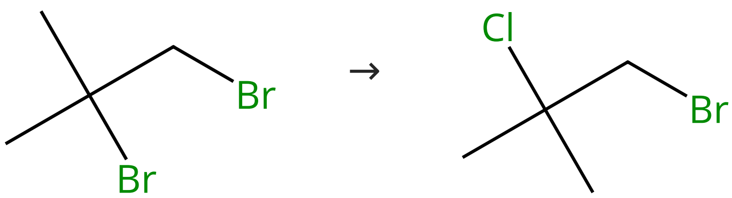
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (44)

Suppliers (13)

31-046-CAS-12082840

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 16 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (36)

Suppliers (53)

31-046-CAS-10571789

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

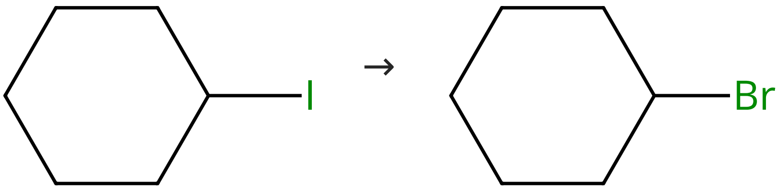
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 17 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (72)

Suppliers (74)

31-046-CAS-6020010

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

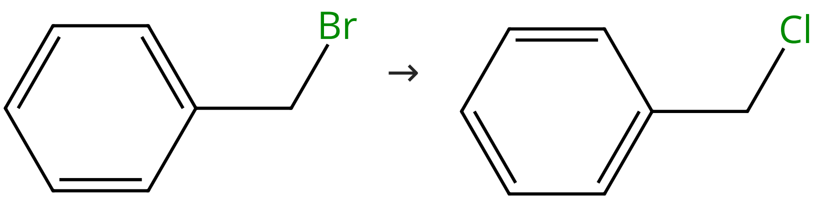
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 18 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (67)

Suppliers (65)

31-046-CAS-5702155

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

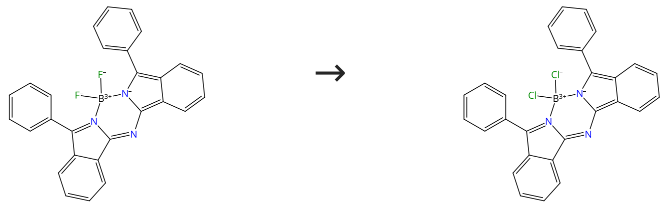
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 19 (1 Reaction)

Steps: **1** Yield: **100%**



31-046-CAS-21792104

Steps: **1** Yield: **100%**

1.1 **Reagents:** Boron trichloride
Solvents: Dichloromethane, Water; 20 min, rt

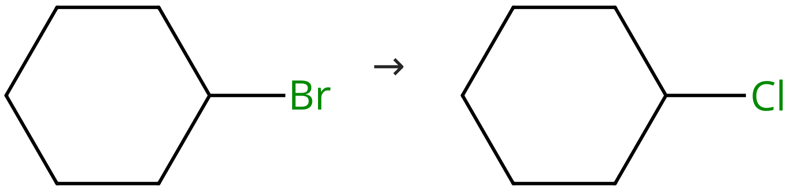
Synthesis, properties and reactivity of B Cl_2 aza-BODIPY complexes and salts of the aza-dipyrrinato scaffold

By: Diaz-Rodriguez, Roberto M.; et al

Organic & Biomolecular Chemistry (2020), 18(11), 2139-2147.

Scheme 20 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (74)

Suppliers (72)

31-046-CAS-8154800

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

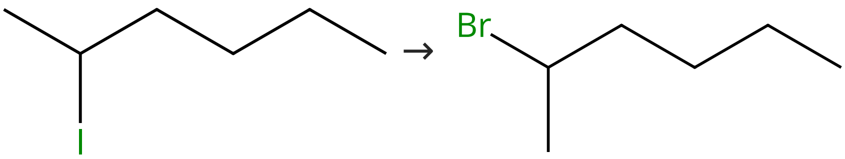
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (15)

Suppliers (52)

31-046-CAS-10283632

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth bromide
Solvents: 1,2-Dichloroethane

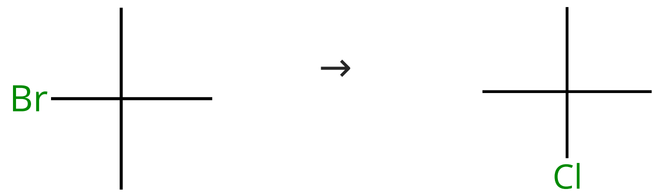
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (60)

Suppliers (61)

31-046-CAS-1159952

Steps: **1** Yield: **100%**

1.1 **Reagents:** Bismuth trichloride
Solvents: 1,2-Dichloroethane

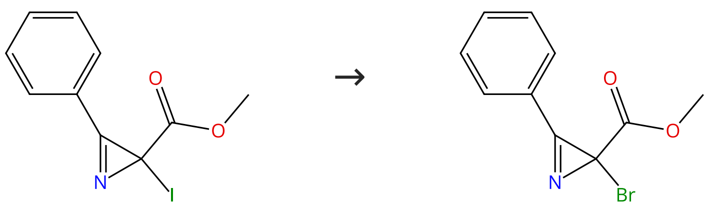
BiX_3 as an efficient and selective reagent for the halogen exchange reaction

By: Boyer, Bernard; et al

Tetrahedron (1999), 55(7), 1971-1976.

Scheme 23 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (2)

31-046-CAS-20895826

Steps: **1** Yield: **99%**

1.1 **Reagents:** Tetrabutylammonium bromide
Solvents: Dichloromethane; 15 h, rt

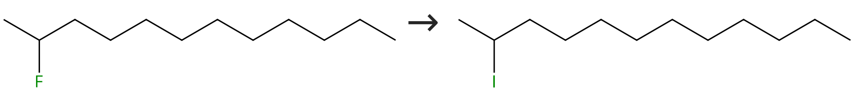
Easy Access to 2-Fluoro- and 2-Iodo-2H-azirines via the Halex Reaction

By: Agafonova, Anastasiya V.; et al

Synthesis (2019), 51(24), 4582-4589.

Scheme 24 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (5)

Supplier (1)

31-046-CAS-8172206

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Solvents: Hexane; 90 min, rt

1.2 rt → -30 °C; 30 min, - 30 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

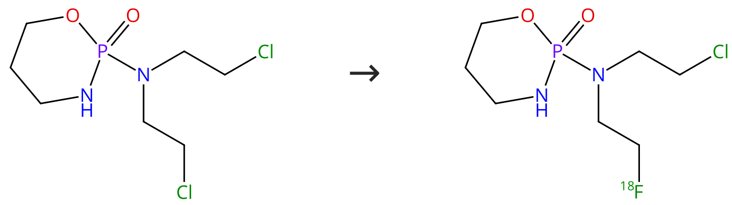
Experimental Protocols

Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 25 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (51)

31-046-CAS-5581608 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium carbonate, [^{18}*F*]-Fluoride
Catalysts: 2,2,2-Cryptand
Solvents: Acetonitrile, Water; rt; 110 °C

1.2 **Solvents:** Acetonitrile; 10 min, 110 °C; 3 min, 110 °C → rt

Synthesis of 2-[(2-chloro-2'-[^{18}F]fluoroethyl)amino]-2H-1,3,2-oxazaphosphorinane-2-oxide (^{18}F-fluorocyclophosphamide), a potential tracer for breast tumor prognostic imaging with P ET

By: Lacan, Goran; et al

Journal of Labelled Compounds & Radiopharmaceuticals (2005), 48(9), 635-643.

Scheme 26 (1 Reaction) Steps: **1** Yield: **99%**

31-046-CAS-13465248 Steps: **1** Yield: **99%**

1.1 **Reagents:** Bromide ($^{76}Br^{1-}$)
Solvents: Acetonitrile, Water; 100 °C

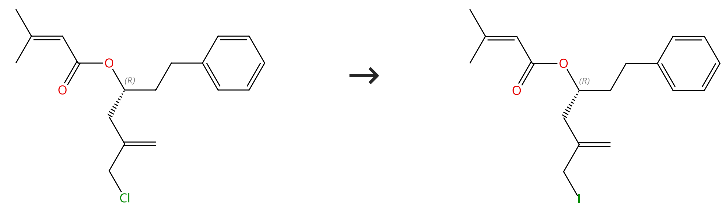
1.2 **Catalysts:** *trans*-Di(μ-acetato)bis[*o*-(di-*o*-tolylphosphino) benzyl]dipalladium
Solvents: Toluene; 40 min, 110 °C

Radiobromination of closo-carboranes using palladium-catalyzed halogen exchange

By: Winberg, Karl Johan; et al

Journal of Labelled Compounds & Radiopharmaceuticals (2005), 48(3), 195-202.

Scheme 27 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-8528183 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

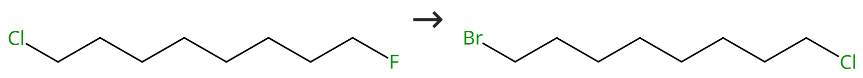
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 28 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (9)

Suppliers (27)

31-046-CAS-3632536

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Catalysts: Titanocene dibromide
Solvents: Hexane; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt

1.2 **Reagents:** Dichloromethane; 24 h, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

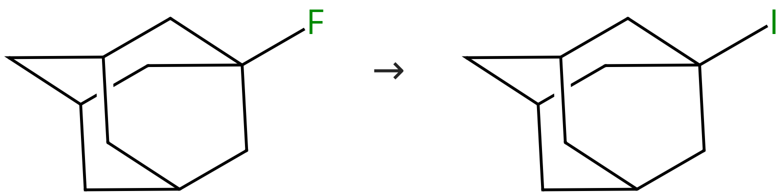
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 29 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (58)

Suppliers (56)

31-046-CAS-14562042

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Solvents: Hexane; 90 min, rt

1.2 rt → -30 °C; 30 min, -30 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

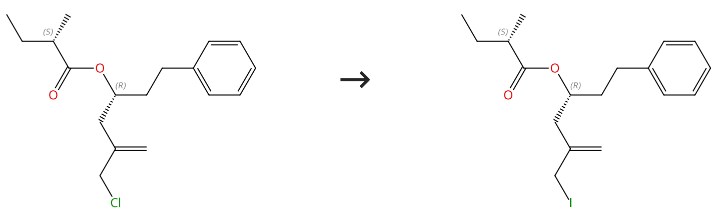
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 30 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-10708336

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

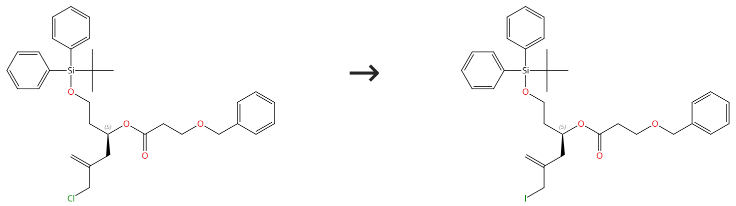
Experimental Protocols

A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 31 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+) → Absolute stereochemistry shown, Rotation (+)

31-046-CAS-3913434 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

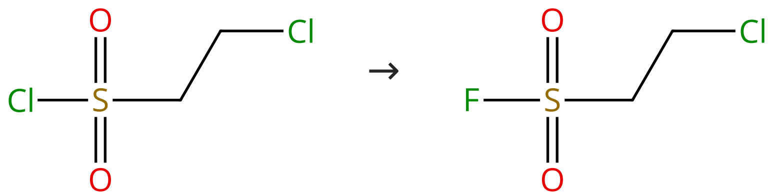
Experimental Protocols

A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 32 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (58) Suppliers (33)

31-046-CAS-21038759 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium bifluoride
Solvents: Water; 2 h, 22 °C

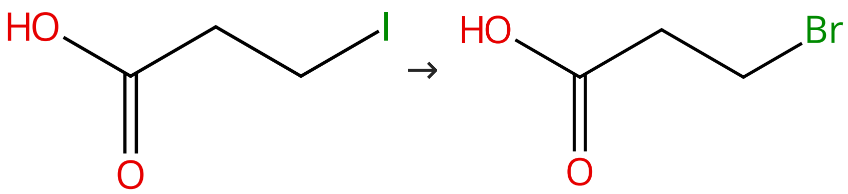
Experimental Protocols

Ethenesulfonyl Fluoride (ESF): An On-Water Procedure for the Kilogram-Scale Preparation

By: Zheng, Qinheng; et al

Journal of Organic Chemistry (2016), 81(22), 11360-11362.

Scheme 33 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (58) Suppliers (54)

31-614-CAS-35771525 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,8-Dibromooctane; 24 h, 130 °C

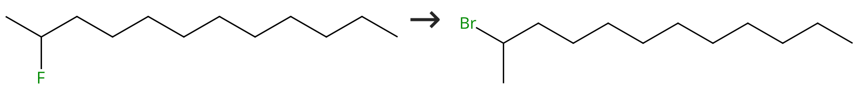
Experimental Protocols

Zeolites catalyze the halogen exchange reaction of alkyl halides

By: Mingueza-Verdejo, Paloma; et al

Catalysis Science & Technology (2023), 13(8), 2308-2316.

Scheme 34 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (5) Suppliers (18)

31-046-CAS-13032525 Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Catalysts: Titanocene dibromide
Solvents: Hexane; 0 °C; 30 min, 0 °C; 30 h, 0 °C → rt; rt

1.2 **Reagents:** Dibromomethane
Solvents: Hexane; rt; 30 min, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

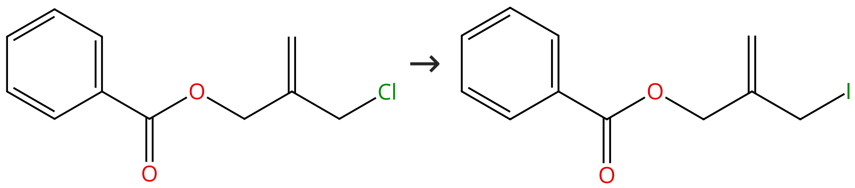
Experimental Protocols

Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 35 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (3) Suppliers (2)

31-046-CAS-9139613 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 16 h, reflux

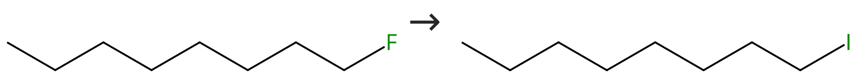
Experimental Protocols

Practical preparation of 2-halomethyl-allyl carboxylates

By: Sun, Chong-Si; et al

Journal of the Chinese Chemical Society (Taipei, Taiwan) (2008), 55(2), 435-438.

Scheme 36 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (41) Suppliers (72)

31-046-CAS-3906334 Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Solvents: Hexane; rt → -40 °C

1.2 **Reagents:** Diiodomethane; -40 °C; -40 °C → rt; 24 h, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

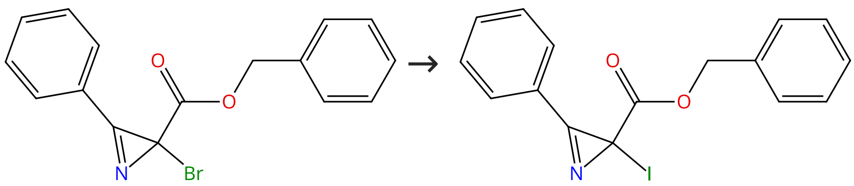
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 37 (1 Reaction)

Steps: **1** Yield: **99%**



31-046-CAS-20895825

Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium iodide
Solvents: Acetone; 48 h, rt

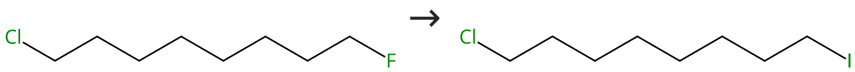
Easy Access to 2-Fluoro- and 2-Iodo-2H-azirines via the Halex Reaction

By: Agafonova, Anastasiya V.; et al

Synthesis (2019), 51(24), 4582-4589.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (9)

Suppliers (3)

31-046-CAS-6035781

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Solvents: Hexane; rt → -40 °C

1.2 **Reagents:** Diiodomethane; -40 °C; -40 °C → rt; 24 h, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

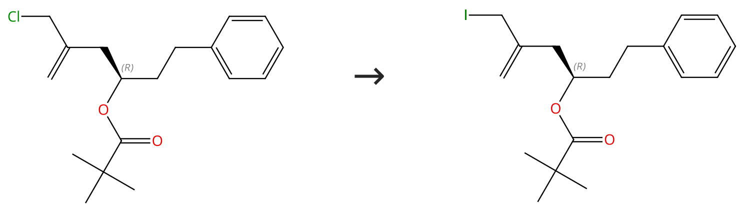
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-3966311

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

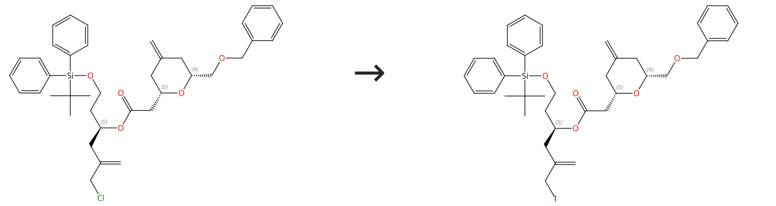
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-046-CAS-1479651

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

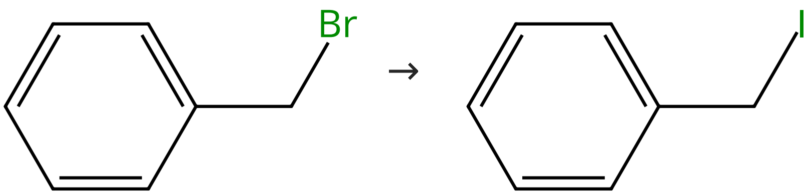
A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 41 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (67)

Suppliers (41)

31-614-CAS-35771530

Steps: **1** Yield: **99%**

1.1 **Reagents:** Butyl iodide; 24 h, 130 °C

Experimental Protocols

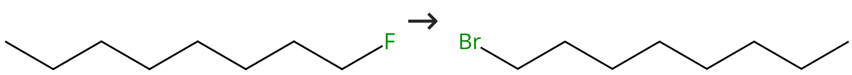
Zeolites catalyze the halogen exchange reaction of alkyl halides

By: Mingueza-Verdejo, Paloma; et al

Catalysis Science & Technology (2023), 13(8), 2308-2316.

Scheme 42 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (41)

Suppliers (79)

31-046-CAS-14892591

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triethylaluminum
Catalysts: Titanocene dibromide
Solvents: Hexane; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt

1.2 **Reagents:** Dichloromethane; 24 h, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

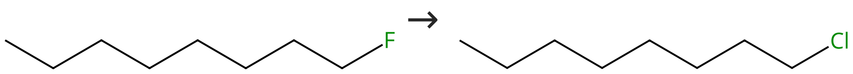
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 43 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (41) Suppliers (56)

31-046-CAS-12763522

Steps: **1** Yield: **99%**

1.1 **Reagents:** Triisobutylaluminum
Catalysts: Titanocene dichloride
Solvents: Hexane; 0 °C; 30 min, 0 °C; 0 °C → rt; 30 min, rt

1.2 **Reagents:** Dichloromethane; 24 h, rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

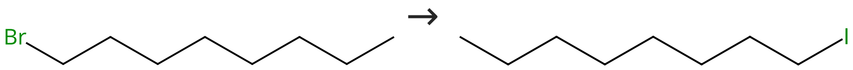
Halogen Exchange Reaction of Aliphatic Fluorine Compounds with Organic Halides as Halogen Source

By: Mizukami, Yuki; et al

Organic Letters (2015), 17(24), 5942-5945.

Scheme 44 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (79) Suppliers (72)

31-046-CAS-9047798

Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium iodide
Catalysts: 1-Hexanaminium, *N*-methyl-*N*,*N*-di-2-propen-1-yl-, iodide (1:1), polymer with *N*,*N*-dimethyl-*N*-2-propen-1-yl-2-propen-1-aminium chloride (1:1) and *N*,*N*'-methylenebis[2-propenamide]
Solvents: Acetonitrile; 12 h, 80 °C

Experimental Protocols

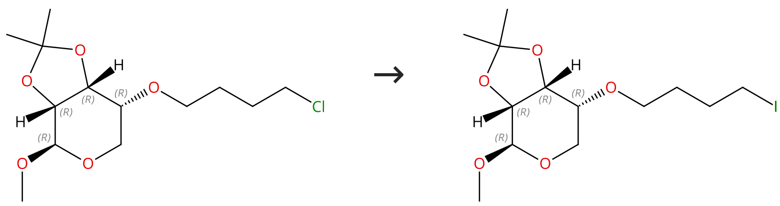
Synthesis and applications of cross-linked poly(diallyldimethyl ammonium chloride) and its derivative copolymers as efficient phase transfer catalyst for nucleophilic substitution reactions

By: Mahdavi, Hossein; et al

Chinese Journal of Polymer Science (2011), 29(2), 165-172.

Scheme 45 (1 Reaction)

Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

31-046-CAS-10912983

Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, reflux

Experimental Protocols

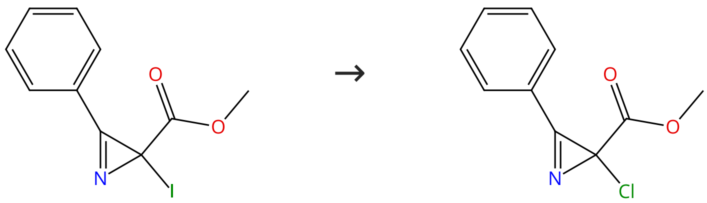
Synthesis and hydrolysis studies of novel glyco- functionalized platinum complexes

By: Moeker, Janina; et al

Carbohydrate Research (2012), 348, 14-26.

Scheme 46 (1 Reaction)

Steps: **1** Yield: **98%**



31-046-CAS-20895828

Steps: **1** Yield: **98%**

1.1 **Reagents:** Tetrabutylammonium chloride
Solvents: Dichloromethane; 6 h, rt

Easy Access to 2-Fluoro- and 2-Iodo-2H-azirines via the Halex Reaction

By: Agafonova, Anastasiya V.; et al

Synthesis (2019), 51(24), 4582-4589.

Scheme 47 (1 Reaction)

Steps: **1** Yield: **98%**

Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

31-046-CAS-12449778

Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium iodide
Solvents: Acetone; 24 h, rt

Experimental Protocols

A New Construction of 2-Alkoxypyrans by an Acylation-Reductive Cyclization Sequence

By: Heumann, Lars V.; et al

Organic Letters (2007), 9(10), 1951-1954.

Scheme 48 (1 Reaction)

Steps: **1** Yield: **98%**

Suppliers (2)

31-046-CAS-20895827

Steps: **1** Yield: **98%**

1.1 **Reagents:** Tetrabutylammonium chloride
Solvents: Dichloromethane; 12 h, rt

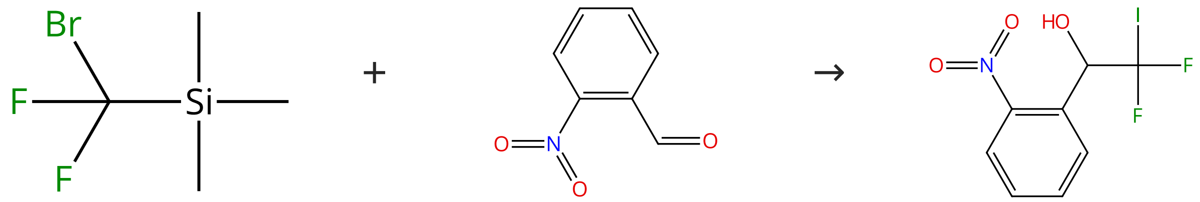
Easy Access to 2-Fluoro- and 2-Iodo-2H-azirines via the Halex Reaction

By: Agafonova, Anastasiya V.; et al

Synthesis (2019), 51(24), 4582-4589.

Scheme 49 (1 Reaction)

Steps: **1** Yield: **97%**



Suppliers (56)

Suppliers (99)

31-046-CAS-1055805	Steps: **1** Yield: **97%**
1.1	**Reagents:** Lithium bromide, Sodium iodide **Solvents:** 1,2-Dimethoxyethane; 1 h, 80 °C; 80 °C → rt
1.2	**Reagents:** Trifluoroacetic acid **Solvents:** Ethanol; 1 h, rt

Experimental Protocols

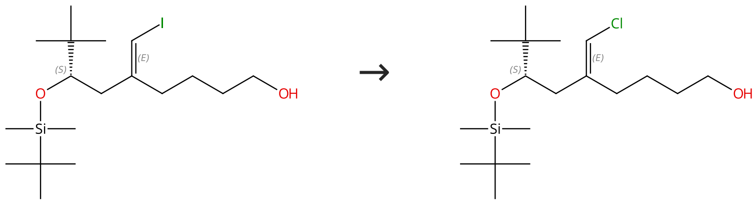
Nucleophilic Iododifluoromethylation of Aldehydes Using Bromine/Iodine Exchange

By: Levin, Vitalij V.; et al

Journal of Organic Chemistry (2015), 80(18), 9349-9353.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (+)
Double bond geometry shown

Absolute stereochemistry shown, Rotation (+)
Double bond geometry shown

31-046-CAS-16205539	Steps: **1** Yield: **97%**
1.1	**Reagents:** Tetramethylammonium chloride **Catalysts:** Cuprous iodide, *trans*-*N*,*N'*-Dimethyl-1,2-cyclohexanediamine **Solvents:** Dimethyl sulfoxide; 96 h, 110 °C

Experimental Protocols

Development of a general copper-catalyzed vinylic Finkelstein reaction-application to the synthesis of the C1-C9 fragment of laingolide B

By: Nitelet, Antoine; et al

Tetrahedron (2016), 72(40), 5972-5987.